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**Adaptive Explainable Ensemble with NLP Fusion for Pre-Launch
Steam Game Reception Prediction under Varying Feature Availability**

Literature Review

Student details:

Name: Heshan Kaushith Ratnaweera

Degree program: Computer Science BSc Honours with Industry Placement

Student Identification: W2082289(UOW) | 20222094(IIT)

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2.1 Introduction

The global video game industry has grown into a major commercial sector, with the market reaching \$300 billion during 2020–2021 (Fan, 2023) and generating an estimated \$249.6 billion in 2023 (Peri, 2024). On the Steam platform, user review scores serve as a primary indicator of player reception and commercial success (Kraemer et al., 2025; Santos et al., 2019). Yet accurately predicting whether a game will achieve positive reception (operationalized here as a $\geq 75\%$ positive review ratio) using only metadata available before release remains an unresolved problem: the post-launch information on which most predictive models depend is unavailable at the point when design and pricing decisions still can be revised.

This chapter draws together research on Steam user reviews, machine learning approaches to game success prediction, and pre-launch prediction methods from adjacent domains. The aim is to evaluate established techniques, identify their limitations, and construct the evidence base that justifies the specific combination of methods proposed in this study.

Seven thematic areas are examined: the reliability of Steam user reviews as quality signals; tabular ensemble classifiers for binary prediction tasks; cross-domain pre-launch prediction evidence; hybrid models combining structured features with pre-trained language model embeddings; class imbalance handling techniques; adaptive confidence-based routing mechanisms; and post-hoc interpretability methods. The review draws primarily on work published between 2020 and 2026, with foundational contributions included where they establish the basis for more recent developments.

Section 2.2 outlines background concepts and fundamental theories. Section 2.3 presents a thematic review of the literature. Section 2.4 offers a critical synthesis across themes. Section 2.5 identifies research gaps, and Section 2.6 provides a chapter summary.

2.2 Background Concepts / Fundamental Theories

This section defines the core concepts underpinning the research.

Pre-launch Metadata

Pre-launch metadata refers to information available about a video game before its public release on platforms such as Steam. This typically includes structured attributes such as genre, price, DLC (downloadable content) count, age rating, content tags, supported platforms, and developer-written textual descriptions (short_description). These features are determined during the design and production phases and do not depend on post-release player behavior or review data (Aleem and Noor, 2024; Sahu et al., 2022).

User Reception and Binary Classification

User reception on Steam is commonly measured through review scores, with positive reception typically defined as 75% or higher positive reviews. For research in this domain, the prediction task is formulated as a binary classification problem: determining whether a game will achieve "well-received" or "not well-received" status based solely on pre-launch features. This binary framing aligns with analogous success prediction tasks across domains (Kraemer et al., 2025; Sahu et al., 2022).

Tabular Data and Ensemble Methods

Tabular data consists of structured rows and columns, typical of game metadata. Tree-based ensemble methods such as XGBoost, Random Forest, and gradient boosting are well-established performers on tabular binary classification tasks, frequently outperforming linear models and deep learning approaches across published benchmarks (Shwartz-Ziv and Armon, 2021; Grinsztajn et al., 2022; Uddin and Lu, 2024).

Hybrid NLP-Tabular Fusion

Hybrid models combine structured tabular features with textual information processed through pre-trained language models. BERT (Bidirectional Encoder Representations from Transformers; Devlin et al., 2019) introduced the transformer-based pre-training paradigm that underpins modern NLP embeddings. Sentence-BERT (Reimers and Gurevych, 2019) extends this to produce fixed-length sentence embeddings that can be fused with tabular data via concatenation or more advanced mechanisms. Such fusion has produced performance gains in domains with both

structured and textual pre-launch information (Gunduz, 2024; Kasneci and Kasneci, 2024; Gameiro, 2025; Qin et al., 2023).

Class Imbalance

Class imbalance occurs when one class (e.g., well-received games) substantially outnumbers the other, which is common in reception prediction tasks. Techniques to address this include oversampling via SMOTE (Synthetic Minority Oversampling Technique; Chawla et al., 2002), class weighting, and ensemble methods (Abdelhamid and Desai, 2024; Carvalho et al., 2025).

Post-hoc Explainability (SHAP and LIME)

Explainability methods allow practitioners to interpret black-box model predictions. SHAP (SHapley Additive exPlanations) provides consistent feature attribution grounded in game theory (Lundberg and Lee, 2017), while LIME (Local Interpretable Model-agnostic Explanations) fits a simple surrogate model locally around individual predictions (Ribeiro et al., 2016). Both are model-agnostic and applicable to tabular and hybrid models (Yasui and Sato, 2025; Salih et al., 2023).

Adaptive Architectures and Confidence-based Routing

Adaptive systems route predictions to different specialist models depending on available features or prediction confidence. Confidence is typically measured using the maximum predicted probability from a model's output (Alajmi and Pergola, 2025/2026; Wang et al., 2024). Applied across graduated feature tiers, this enables the system to generate predictions at different development stages without requiring a complete feature set.

Feature Engineering and Derived Features

Feature engineering involves constructing new variables from raw data, such as interaction terms (e.g., price-to-DLC ratio) or domain-specific indices. This process can yield material performance improvements in structured prediction tasks (Sahu et al., 2022; Tschalzev et al., 2024).

These concepts collectively underpin the literature examined in the sections that follow.

2.3 Thematic Literature Review

This section organizes the literature into key themes relevant to the development of an adaptive explainable ensemble model for pre-launch Steam game reception prediction.

2.3.1 Reliability of Steam User Reviews as Quality Signals

Steam user reviews occupy an analytically awkward position: they are commercially meaningful yet the evidence consistently questions their reliability as neutral quality indicators. Kraemer et al. (2025) confirm that both user and critic review scores positively influence game sales, but establish a meaningful asymmetry between them: critic reviews exert greater market influence than user reviews, even though user scores are more plentiful and publicly accessible. Santos et al. (2019) offer a complementary but more critical perspective, finding that user scores are more polarised and more susceptible to strategic manipulation than critic assessments. Taken together, these two studies converge on the same conclusion through different routes: user reviews are commercially consequential, but structurally less reliable than expert evaluations.

This picture is complicated further by genre and context. Busurkina et al. (2020) find that the relationship between user sentiment and expert assessment varies substantially across game genres, which means that the reliability of Steam reviews as a quality proxy is not uniform across the platform. Li et al. (2021) add a further qualification: user reviews are most accurate at capturing concrete, playability-related issues and weakest as holistic quality measures. The implication is not that review scores are unusable, but that their meaning is context-dependent. Where Santos et al. and Kraemer et al. identify structural biases at the aggregate level, Busurkina et al. and Li et al. show that these biases manifest differently depending on game type and reviewer motivation.

For prediction purposes, these limitations collectively argue in favor of a conservative, well-established operationalization of reception quality. The $\geq 75\%$ positive review threshold applied in this study reflects that reasoning: it is cautious enough to discount moderate, ambiguous scores and well-established enough to have been adopted as a benchmark in prior work on Steam reception (Kraemer et al., 2025).

2.3.2 Tabular Ensemble Methods for Binary Classification

The superiority of tree-based ensemble methods over alternative classifiers on structured tabular data is one of the most consistently replicated findings in recent machine learning benchmarking literature. Shwartz-Ziv and Armon (2021), Grinsztajn et al. (2022), and Uddin and Lu (2024) each investigate this question through different empirical lenses (benchmark datasets, synthetic perturbation analysis, and statistical significance testing respectively) and arrive at the same conclusion: XGBoost (Chen and Guestrin, 2016), Random Forest (Breiman, 2001), and gradient boosting variants outperform linear models, SVMs, k-NN, and many deep learning approaches across a broad range of tabular binary classification tasks. Bentéjac et al. (2019) reinforce this through a comparative analysis of gradient boosting algorithms specifically, confirming consistent performance across diverse settings.

This consensus carries an important qualification that is often understated. Grinsztajn et al. (2022) acknowledge that the degree of outperformance varies with dataset characteristics, particularly the proportion of uninformative features and the underlying data-generating distribution. Shwartz-Ziv and Armon (2021) make a parallel observation: the advantage of tree ensembles is not absolute across all tabular configurations. These caveats matter directly for the present study because game metadata across different development tiers varies considerably in both feature count and information density. The practical implication is that no single classifier should be assumed optimal across all feature configurations, which motivates an adaptive, multi-tier architecture rather than the application of a single fixed model to all inputs regardless of their completeness.

While the weight of evidence strongly endorses tree ensembles as a backbone classifier for tabular data, the dataset-dependent variation acknowledged by Grinsztajn et al. (2022) underscores that empirical comparison across multiple candidate classifiers, rather than prescriptive selection of a single algorithm, is the methodologically appropriate approach when the input distribution is heterogeneous across tiers.

2.3.3 Pre-launch Prediction in Cross-Domain Studies

Pre-launch prediction has been addressed across several structurally comparable domains, and the evidence from these studies is more informative when read in aggregate than when examined

individually. A recurrent methodological pattern emerges: each study isolates pre-launch features, applies engineered or derived variables, and evaluates a binary or categorical success outcome. What varies across domains is the relative contribution of feature engineering, model architecture, and textual modality; this variation yields transferable lessons for the gaming context.

In film, Sahu et al. (2022) report that derived feature engineering contributes a 17.62% accuracy improvement over base models, achieving 96% classification accuracy on pre-production metadata. This figure should be interpreted carefully: the result is domain-specific and the architecture is complex, which limits direct replication. Nevertheless, it establishes the principle that engineered variables can carry substantial predictive weight in pre-launch classification tasks. Aleem and Noor (2024) take a simpler approach in the Android app domain, showing that pre-launch structured attributes alone (category, price, and rating) are sufficient to support effective prediction using Random Forest, SVM, XGBoost, and k-NN. Where Sahu et al. show that engineering beyond raw features adds material value, Aleem and Noor suggest that even minimal structured metadata can sustain classification performance. The two studies are not directly comparable given their differences in domain complexity and feature richness, but together they indicate that structured pre-launch metadata is both necessary and improvable through purposeful feature construction.

On the representational side, Gunduz (2024) provides a directly relevant comparison: BERT representations of creator-written campaign text outperform FastText embeddings for crowdfunding outcome prediction, establishing that deep contextual encodings extract more signal from short-form pre-launch prose than simpler word-based methods. This finding stands in productive tension with Alajmi and Pergola (2025/2026), who show that confidence-based routing between specialist classifiers improves robustness in multi-class NLP tasks. Gunduz and Alajmi and Pergola address different layers of the same system: Gunduz argues for richer text representation, while Alajmi and Pergola argue for adaptive model selection. Neither paper combines both insights within a single unified framework.

Critically, no reviewed cross-domain study combines tiered specialist models, derived feature engineering, NLP fusion, and confidence-based routing within a single unified adaptive framework

applied to gaming reception. Each study addresses one or two of these components in isolation, leaving their combination as an unaddressed methodological gap that the present study directly targets.

2.3.4 Hybrid Models: Integrating NLP with Structured Tabular Features

Hybrid architectures consistently outperform single-modality baselines when both structured and textual data are available, and this finding is replicated across multiple domains and fusion strategies. What the literature does not adequately address is why the improvement occurs and under what conditions different fusion mechanisms work best, which matters for any system choosing between architectural alternatives.

Qin et al. (2023) report a performance gain of nearly 24% over single-modality baselines by fusing transformer-based text features with structured movie metadata, providing one of the most quantified demonstrations of hybrid superiority in the pre-launch prediction literature. Kasneci and Kasneci (2024) offer a complementary perspective: in their ablation study, LLM-derived embedding features produced by concatenating RoBERTa and GPT-2 representations with tabular inputs rank among the most influential contributors to XGBoost and CatBoost performance on imbalanced datasets. These two studies converge on the conclusion that textual embeddings improve performance substantially, but both default to concatenation as the fusion mechanism without comparing it against alternatives.

Gameiro (2025) introduces an important architectural contrast. Rather than simple concatenation, the study applies a synergistic gated fusion layer to combine Sentence-BERT embeddings with structured features, achieving superior Macro F1 scores over concatenation-only baselines. This result raises a question that Qin et al. and Kasneci and Kasneci do not address: whether the consistent performance gains attributed to hybrid fusion reflect the value of the textual modality, the quality of the fusion mechanism, or both. The field has not yet established a rigorous comparison of fusion strategies, which means the relative contribution of concatenation versus gated or attention-based alternatives remains an open empirical question.

The question of interpretability adds a further dimension. Maarouf et al. (2025) apply SHAP to a fused architecture combining LLM embeddings with financial features, aggregating attribution values across embedding dimensions to produce modality-level contribution scores rather than per-dimension decompositions. Kasneci and Kasneci (2024) corroborate this by showing that individual modality contributions remain distinguishable even in high-dimensional fused feature spaces. These findings suggest that hybrid systems need not sacrifice interpretability for performance. For the gaming domain, the Steam short_description field (a fixed-length developer-written text input) represents a practically available and analytically tractable modality for hybrid NLP-tabular fusion, particularly when paired with post-hoc explainability methods capable of attributing predictions across both structured and embedded features.

2.3.5 Class Imbalance Handling and Adaptive Ensemble Techniques

Class imbalance in reception prediction datasets creates a methodological challenge that is easier to recognize than to resolve. Abdelhamid and Desai (2024) and Carvalho et al. (2025) each survey the main categories of response: resampling (including SMOTE), class weighting, cost-sensitive learning, and hybrid strategies. Both arrive at a broadly consistent conclusion: no single method is universally superior, and the relative performance of each technique is dataset-dependent. This absence of a settled answer is itself analytically informative: it argues against prescribing any specific imbalance-handling approach before domain-specific evaluation and in favor of empirically comparing multiple candidate strategies within the modelling pipeline.

The aggregation strategy for multi-model systems introduces a related consideration. Stacking ensembles, which train a secondary meta-learner on the probability outputs of multiple base classifiers, have shown performance benefits in analogous prediction tasks (Liao et al., 2022; Şencan et al., 2026). This mechanism is relevant to the present study because the five-tier architecture requires a principled means of combining predictions across specialist models. However, the stacking studies reviewed by Liao et al. and Şencan et al. operate under the assumption of full feature availability and do not examine how stacking behaves when the feature sets available to constituent classifiers differ substantially across inputs. This gap between established stacking methodology and the variable-feature reality of tiered game development prediction is precisely where confidence-based routing offers a more appropriate alternative.

2.3.6 Adaptive ML and Confidence-Based Routing

Confidence-based routing offers a principled departure from the fixed-model and rigid rule-based selection strategies that characterise most existing adaptive systems. Rather than assigning inputs to a pre-specified model, the approach directs predictions to whichever specialist classifier produces the highest predicted probability (`predict_proba`) among those eligible for the current feature set. Alajmi and Pergola (2025/2026) apply this mechanism in a multi-class NLP task, showing that routing through the most confident available specialist improves robustness compared to fixed assignment. Wang et al. (2024) provide a parallel application in clinical NLP, where confidence-weighted model selection across heterogeneous input conditions yields more stable predictions than any single model applied uniformly. What these two studies share is the underlying principle: when model suitability varies with input characteristics, dynamic confidence-based selection is preferable to static assignment.

A structurally related insight comes from Le Morvan et al. (2020), who show that architectures designed to handle missing values natively outperform imputation-based baselines. This finding is relevant because it establishes the broader principle that systems operating under varying input conditions should accommodate incomplete data structurally rather than through pre-processing corrections. Confidence-based routing applies the same logic at the architecture level: rather than imputing missing features to enable a fixed model, the system adapts its model selection to whatever features are actually available.

What neither Alajmi and Pergola nor Wang et al. address, however, is the degree to which confidence scores remain well-calibrated across structurally different feature tiers. In NLP and clinical settings, all inputs share the same high-level representational space. In a tiered gaming-domain system, the feature vectors available to each specialist model differ not just in quantity but in type and information density. Whether confidence scores serve as a reliable selection criterion under these conditions is an open empirical question that the present study addresses directly.

2.3.7 Post-hoc Explainability Methods

Post-hoc explainability has become a standard complement to high-performing but opaque classifiers, yet the literature reveals an important asymmetry between individual method validation and combined application in hybrid systems. SHAP, introduced by Lundberg and Lee (2017) as a game-theoretic framework for consistent feature attribution, and LIME, proposed by Ribeiro et al. (2016) as a local surrogate explanation method, are each well-supported individually. Recent work by Yasui and Sato (2025) improves LIME's local fidelity through a conditional VAE-based sampling strategy, while Salih et al. (2023) compare both methods systematically across tabular and hybrid settings, confirming that each offers distinct advantages depending on whether global or instance-level attribution is required.

The question of joint application to hybrid NLP-tabular architectures is less settled. SHAP's global attribution and LIME's instance-level surrogate explanations are methodologically complementary in principle, but reconciling them in systems where part of the feature space consists of high-dimensional embedding vectors raises practical challenges. Maarouf et al. (2025) address this in the financial domain by aggregating SHAP values across embedding dimensions into a single modality-level score, which preserves interpretability without requiring per-dimension attribution across potentially hundreds of embedding features. This solution is pragmatically sound, but it represents a design choice with interpretive consequences: it sacrifices fine-grained embedding attribution in favor of modality-level clarity. The trade-off is not examined rigorously in any reviewed study, and the conditions under which per-dimension versus aggregated attribution is preferable for developer-facing decision support remain unaddressed. For the present system, which combines structured tabular features with 50-dimensional PCA-reduced SBERT embeddings, this design decision warrants explicit treatment.

2.4 Critical Analysis and Synthesis

The thematic review in Section 2.3 establishes that each component of the proposed system (tree-based ensemble classification, hybrid NLP-tabular fusion, confidence-based routing, and post-hoc explainability) has been validated in its own domain. What that review also makes clear is that this validation has occurred in isolation: the literature's knowledge is modular, not integrated. Understanding why that modularity is a problem, and what it implies for the present study's design, is the purpose of this section.

The most consistent finding across the reviewed body of work is the performance advantage of tree-based ensemble methods on structured tabular data (Shwartz-Ziv and Armon, 2021; Grinsztajn et al., 2022; Uddin and Lu, 2024). This consensus holds across methodologically diverse benchmarking studies, but it coexists with evidence that the advantage is dataset-dependent and not absolute (Grinsztajn et al., 2022; Shwartz-Ziv and Armon, 2021). Ensemble methods therefore provide a well-justified starting point for any tabular classifier, while the specific algorithm choice should remain empirically determined rather than prescribed in advance.

A second area of agreement, reinforced by multiple cross-domain studies, is that hybrid NLP-tabular architectures consistently outperform single-modality baselines when creator-written text is available (Qin et al., 2023; Kasneci and Kasneci, 2024; Gameiro, 2025; Gunduz, 2024). This consensus, however, conceals an unexamined question: because most hybrid studies rely on simple concatenation without comparing alternative fusion strategies, it remains unclear whether performance gains are attributable to the textual modality, the fusion mechanism, or their interaction. Gameiro (2025) provides a partial exception through a comparison of gated and concatenation-based fusion, but no reviewed study offers a systematic multi-strategy evaluation. This constitutes both a limitation in the existing evidence and a design question the present study can address empirically.

The more fundamental tension in the reviewed literature, however, is not between competing methods but between the questions the literature asks and the questions practitioners need answered. Gaming-domain research is predominantly post-launch in orientation: it analyses review sentiment (Santos et al., 2019; Busurkina et al., 2020), player retention (Mulla et al., 2025), and concurrent user counts (Wirawan and Kusuma, 2024), all of which are unavailable before a game ships. Cross-domain studies in film (Sahu et al., 2022), crowdfunding (Gunduz, 2024), and app stores (Aleem and Noor, 2024) have addressed structurally identical pre-launch prediction problems, yet their findings have not been adapted to the gaming context in any reviewed work. This asymmetry between what adjacent fields have established and what the gaming literature has adopted is the deepest and most practically consequential gap in the evidence base.

Explainability methods occupy a similarly disconnected position. SHAP and LIME are individually well-validated (Lundberg and Lee, 2017; Ribeiro et al., 2016; Yasui and Sato, 2025; Salih et al., 2023) but have not been applied jointly to adaptive multi-model systems combining structured and embedded features. The few studies that apply post-hoc attribution to hybrid systems, such as Maarouf et al. (2025), operate in financial rather than gaming contexts and do not address how attribution behaves when a routing architecture introduces variability in which model produces each prediction. For a system that selects between five specialist models at prediction time, this is not a minor technical consideration: the explainability mechanism must remain coherent across routing decisions.

Taken together, these observations point to a consistent pattern in the reviewed literature. The field has developed well-validated components but has not attempted their integration into a system that is simultaneously adaptive, hybrid, explainable, and pre-launch in orientation. The gaps identified in Section 2.5 are therefore not a collection of isolated omissions; they reflect a structural characteristic of a literature that has studied parts without examining the whole.

2.5 Research Gap Analysis

The preceding review identifies several distinct gaps that, taken together, define the rationale for the proposed adaptive explainable ensemble with NLP fusion for pre-launch Steam game reception prediction.

Practical Gap

Despite the commercial importance of user reception on Steam (Kraemer et al., 2025; Peri, 2024), existing studies in the gaming domain predominantly rely on post-launch metrics such as review counts, peak concurrent users, player activity, and engagement data (Ma et al., 2025; Ellington, Pramanik and Robinson, 2026; Wirawan and Kusuma, 2024; Hu et al., 2024; Teja et al., 2023). No reviewed work in the gaming domain formulates the problem as a true pre-launch binary classification task using only metadata and developer-written descriptions available before release. Developers therefore lack evidence-based tools to assess reception risk during design, pricing, and production phases when interventions remain feasible.

Methodological Gap

While tree-based ensembles perform reliably on tabular data (Shwartz-Ziv and Armon, 2021; Grinsztajn et al., 2022; Uddin and Lu, 2024) and hybrid NLP-tabular models show promise in other domains (Qin et al., 2023; Kasneci and Kasneci, 2024; Gameiro, 2025; Gunduz, 2024), these approaches have not been systematically combined or adapted for the gaming context. Most gaming studies do not explore derived feature engineering (Sahu et al., 2022), confidence-based model routing (Alajmi and Pergola, 2025/2026; Wang et al., 2024), or specialist models trained on graduated feature tiers that reflect progressive metadata availability during development.

Evaluation and Data Gap

There is a notable absence of labelled benchmarks pairing pre-launch Steam metadata with binary reception outcomes ($\geq 75\%$ positive reviews). Existing game-related research tends to address regression, popularity prediction, revenue estimation, or churn rather than binary reception quality classification (Ellington, Pramanik and Robinson, 2026; Wirawan and Kusuma, 2024; Mulla et al., 2025). Few studies address class imbalance handling specifically within pre-launch reception prediction (Abdelhamid and Desai, 2024; Carvalho et al., 2025), and post-hoc explainability is rarely integrated into adaptive multi-model systems for this domain.

Deployment Gap

Current systems assume a complete, fixed feature set at prediction time. None explicitly supports varying feature availability across development stages (concept to production to pre-release), and none combines confidence-based routing with dual SHAP and LIME explainability for developer decision support (Yasui and Sato, 2025; Salih et al., 2023; Alajmi and Pergola, 2025/2026).

Description Gap

No reviewed study in the gaming domain has examined developer-written text (`short_description`) as a predictive feature for game reception. In adjacent domains, creator-written descriptions have shown strong predictive value for campaign and product success when processed through transformer-based models (Gunduz, 2024; Kasneci and Kasneci, 2024; Gameiro, 2025). The absence of text-based modelling in gaming reception research constitutes a material gap, given

both the widespread availability of developer-written descriptions on Steam and the established predictive value of such text in adjacent domains.

Routing Gap

Existing gaming-domain prediction systems use either a single fixed model or rigid rule-based selection between models. No reviewed study employs confidence-based routing, whereby the system dynamically selects the most confident eligible specialist model at prediction time using predicted probability (Alajmi and Pergola, 2025/2026; Wang et al., 2024). This architectural gap is consequential: confidence-based routing directly enables graceful degradation under partial feature availability, which is a core requirement when predicting reception at different stages of game development (concept, production, pre-release).

Synthesis of Gaps

Cross-domain studies establish the viability of pre-launch prediction, hybrid NLP-tabular fusion, derived feature engineering, and adaptive routing, yet these insights remain unapplied to Steam game reception within a unified and explainable framework. Addressing this gap requires a system that combines adaptive specialist ensembles operating across graduated feature tiers, developer-written text as a predictive modality, confidence-based routing, and integrated post-hoc explainability: a combination that no reviewed study currently provides.

2.6 Chapter Summary

The reviewed literature establishes a detailed picture of what is known and what remains unexamined. Three broad conclusions emerge.

First, the methodological building blocks for pre-launch Steam game reception prediction exist and are well-supported individually. Tree-based ensemble classifiers have shown reliable performance on structured tabular data across diverse benchmarking studies (Shwartz-Ziv and Armon, 2021; Grinsztajn et al., 2022; Uddin and Lu, 2024). Hybrid architectures incorporating Sentence-BERT embeddings with structured features consistently produce performance gains over single-modality baselines in analogous domains (Qin et al., 2023; Kasneci and Kasneci, 2024; Gameiro, 2025; Gunduz, 2024). Derived feature engineering adds material predictive value in pre-

launch classification tasks (Sahu et al., 2022), and both SHAP and LIME provide actionable attribution in tabular and hybrid settings (Lundberg and Lee, 2017; Yasui and Sato, 2025; Salih et al., 2023).

Second, the gaming-domain literature has not adopted these advances. Existing Steam research is predominantly post-launch in orientation, relying on review counts, concurrent player statistics, and engagement metrics unavailable during development (Ma et al., 2025; Ellington, Pramanik and Robinson, 2026; Wirawan and Kusuma, 2024). User review scores, while commercially meaningful (Kraemer et al., 2025), carry documented biases relating to genre, polarization, and review manipulation that complicate their uncritical use as prediction targets (Santos et al., 2019; Busurkina et al., 2020). No reviewed gaming study addresses the full pre-launch prediction problem: formulating it as a binary classification task, handling varying feature availability across development stages, and providing interpretable outputs to guide developer decisions.

Third, the gaps are not merely additive. They reflect a structural asymmetry: cross-domain research has established individually that tiered specialist models, hybrid NLP fusion, confidence-based routing, derived feature engineering, and post-hoc explainability each contribute meaningfully to prediction performance, but none of these studies has combined all components within a unified system applied to Steam game reception prediction. Chapter 3 operationalizes the approach that addresses this gap directly.

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